Gadget House – Product Requirements Document

Product: gadgethouse.in – Storefront, Admin Portal and POS Billing **Version:** 1.0 (Draft)
Date: 13 Nov 2026 **Owner:** Gadget House team **Status:** For review

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1. Overview

Gadget House is an online store for phone accessories: back cases, tempered glass, camera protectors, chargers and smart gadgets. The current site is India-focused (Razorpay, INR, iPhone-heavy catalogue). This PRD defines version 2: a **global-ready storefront**, a secure **Admin Portal** to manage products, orders, staff and access, and a built-in **POS (Point of Sale)** module so the same inventory serves online and in-person sales.

1.1 Problem statement

- Products, stock and prices are hard to manage without a proper admin tool.
- Online and offline (shop counter) sales are tracked separately, so stock goes out of sync.
- The catalogue is organised around a single brand and does not scale to many phone brands and models.
- The store cannot yet serve international customers (currency, language, shipping, taxes, payments).

1.2 Goals

1. Let customers find accessories that fit their exact phone model in under 3 clicks.
2. Let staff add and edit products, variants and stock without a developer.
3. Give the owner full control over who can access what (role-based access).
4. Let staff create GST-compliant POS bills from the admin portal with one button, with stock updating instantly across online and offline.
5. Support international customers: multi-currency, multi-language (incl. RTL), global payments and shipping.

1.3 Non-goals (v1)

- Native mobile apps (PWA is in scope).
- Marketplace / multi-vendor selling.
- Full accounting software (we export data to accounting tools instead).
- Dropshipping automation.

1.4 Success metrics

Metric	Target (6 months after launch)
Storefront conversion rate	⩾ 2.5%
Mobile Lighthouse performance score	⩾ 85
Time to add a new product (with variants)	< 3 minutes
POS bill creation time (3 items)	< 30 seconds
Stock mismatch between system and shelf	< 2%
Checkout abandonment	< 65%
International orders share	⩾ 15%

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2. Users and personas

Persona	Description	Key needs
Shopper (domestic / international)	Buys cases, glass and chargers for their phone	Fit guarantee, fast checkout, trusted payment, local currency
Owner / Super Admin	Runs the business	Full control, sales reports, staff access control, audit trail
Manager	Handles catalogue, offers and orders	Product management, discounts, order handling, reports
Inventory staff	Receives and counts stock	Stock updates, purchase entries, low-stock alerts

Persona	Description	Key needs
Cashier (POS)	Serves walk-in customers	Fast billing, barcode scanning, print/share invoice
Support agent	Handles customer queries	View orders, issue refunds (limited), reply to customers

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3. Scope summary

Module	In v1
Customer storefront (web + PWA)	Yes
Admin Portal (products, orders, customers, content, reports)	Yes
Role-based access control and staff management	Yes
POS billing (launched by a button in Admin)	Yes
Inventory and stock management	Yes
Payments (India + international)	Yes
Shipping and tax configuration	Yes
Notifications (email, WhatsApp, SMS)	Yes
Native apps, loyalty program, marketplace	Later

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4. Storefront requirements (gadgethouse.in)

4.1 Global layout

- **Announcement bar:** rotating messages (free shipping threshold, secure payments, easy returns). Content is editable from admin.
- **Sticky header:** logo, search bar with suggestions, country / language / currency selector, dark/light toggle, sign in, wishlist, cart with count badge.

- **Category navigation:** *Shop by Phone* mega-menu (brand → model), All Products, Back Cases, Tempered Glass, Smart Gadgets, Accessories, Camera Protectors.
- **Footer:** shipping and returns, track order, FAQ, contact, payment icons, legal links, language/currency selector.
- **Floating widgets:** PWA install prompt (dismissible, shown after engagement), WhatsApp / chat button.

4.2 Homepage sections

1. Hero carousel (4–5 slides, managed from admin: image, headline, subtitle, CTA link, schedule dates).
2. **Shop by Phone** split card: brand tabs, model search, model grid, "Show all models".
3. Trust strip (global shipping, secure checkout, easy returns, support).
4. Category grid.
5. Best sellers and new arrivals carousels.
6. Offer banner with countdown timer.
7. Customer reviews (verified buyers only).
8. Newsletter signup and social links.

4.3 Shop by Phone (core feature)

- Brand-first selection (Apple, Samsung, Google Pixel, OnePlus, Xiaomi, Nothing, Motorola, and more added from admin), then model.
- Typing in the model search filters instantly (fuzzy match: "17 pro" finds "iPhone 17 Pro").
- Selecting a model shows **only compatible** products, with a persistent "My phone: iPhone 17 Pro ×" chip across the site.
- The selected phone is remembered (local storage, and account if signed in).
- Product cards and detail pages show a clear **"Fits your phone"** badge or a warning if not compatible.

4.4 Catalogue and product pages

- Listing pages: filters (brand, model, category, price, colour, material, rating, in stock, MagSafe), sorting, pagination or infinite scroll, skeleton loaders.

- Product detail page: image gallery with zoom and video, variants (model, colour, finish), price in selected currency, strikethrough price and discount badge, stock status, delivery estimate by PIN / postal code, compatible-model list, specs, reviews, Q&A, related items, "Frequently bought together".
- Wishlist, recently viewed, and back-in-stock notification.

4.5 Cart and checkout

- Cart drawer with quantity edit, coupon code, shipping and tax estimate shown early.
- Guest checkout plus account checkout (email/phone OTP, Google sign-in).
- Address form that adapts by country (PIN code for India, ZIP for US, postcode for UK, and so on) with address validation.
- Shipping method selection with price and ETA.
- Payment: UPI, cards, net banking, wallets, COD (India, configurable by PIN code); Stripe and PayPal for international cards and wallets; Apple Pay and Google Pay.
- Order confirmation page and email with invoice; WhatsApp / SMS updates for Indian customers.

4.6 Customer account

Orders and tracking, returns / exchange requests, saved addresses, saved phone model(s), wishlist, GST invoice download, notification preferences.

4.7 Global readiness

Area	Requirement
Language	Translation-ready UI (English first; Hindi, Arabic, Spanish, French as next set). RTL layout for Arabic / Hebrew. Allow ~30% text expansion.
Currency	Base currency INR; display in USD, EUR, GBP, AED, and others using daily exchange rates; localised number and date formats.
Pricing	Per-market price overrides (optional) and psychological rounding rules.

Area	Requirement
Tax	GST for India (CGST/SGST/IGST, HSN codes). VAT / sales tax rules for exports configurable per country. Tax-inclusive or exclusive display per region.
Shipping	Zones and rates by country, weight-based and flat-rate options, free-shipping thresholds per currency, duties/customs notice for international orders.
Compliance	Privacy policy and cookie consent (GDPR-style), terms, return policy, accessible per country.
Detection	Detect country by IP and suggest currency and language; the user can always override.

4.8 SEO and performance

- Server-side rendering for category and product pages; clean URLs; sitemap.xml and robots.txt.
- Schema.org markup (Product, Offer, Review, Breadcrumb); hreflang tags for languages.
- Image optimisation (WebP/AVIF, responsive sizes, lazy loading), CDN delivery.
- Core Web Vitals targets: LCP < 2.5s, CLS < 0.1, INP < 200ms.
- PWA: installable, offline fallback page, cached shell.

4.9 Accessibility

WCAG 2.1 AA contrast, keyboard navigation, visible focus, alt text on all images, minimum 44px touch targets, screen-reader labels on icon-only buttons.

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5. Admin Portal requirements

The Admin Portal lives at a separate protected route or subdomain (for example `admin.gadgethouse.in`), is responsive (usable on tablet), and supports dark and light mode.

5.1 Navigation (sidebar)

Dashboard · **POS (button)** · Products · Inventory · Orders · Customers · Discounts · Content (banners, pages) · Reports · Staff & Roles · Settings.

The top bar has a prominent **"Open POS"** button (visible only to roles with POS permission), a global search (orders, products, customers), notifications, and the profile menu.

5.2 Dashboard

- Today's sales (online, POS, total), orders to fulfil, low-stock alerts, pending returns.
- Sales chart (7 / 30 / 90 days), top products, top phone models, traffic by country.
- Quick actions: Add product, Create POS bill, Create discount.

5.3 Product management

Add / edit product form

Field group	Fields
Basics	Title, slug, short description, rich-text description, category, subcategory, brand (of accessory), tags, status (draft / active / archived)
Media	Multiple images (drag to reorder, alt text), video link, auto-resize and compress
Compatibility	Select compatible phone brand(s) and model(s) in bulk (checkbox tree)
Variants	Options (colour, finish, model), per-variant SKU, barcode, price, compare-at price, cost price, weight, stock
Pricing and tax	Base price (INR), per-market price overrides, HSN code, GST rate, tax-inclusive toggle
Inventory	Track stock toggle, stock per location (warehouse, shop), low-stock threshold, allow backorder
SEO	Meta title, meta description, URL slug, social image

Field group	Fields
Shipping	Weight, dimensions, shipping class, country restrictions

Capabilities

- Bulk import and export (CSV / Excel) with a downloadable template and validation report.
- Bulk edit (price, status, category, stock).
- Duplicate product; generate variants from options automatically.
- Barcode generation and label printing (SKU / EAN / QR).
- Manage categories, phone brands and models (add a new phone model once; it is available across all products).
- Product change history (who changed what and when).

5.4 Inventory management

- Stock by location (online warehouse, shop counter) with transfers between locations.
- Purchase entries (supplier, invoice number, cost, quantity) that add stock and update cost price.
- Stock adjustment with required reason (damage, loss, recount).
- Low-stock and out-of-stock alerts by email / WhatsApp.
- Stock ledger per SKU showing every movement (sale, POS, return, adjustment, purchase).

5.5 Order management

- Order list with filters (status, channel: online / POS, payment status, country, date) and saved views.
- Order detail: items, customer, addresses, payment, timeline, notes, invoice.
- Statuses: Pending → Paid → Packed → Shipped → Delivered; Cancelled; Returned; Refunded.
- Actions: print invoice and packing slip, create shipment and tracking number (courier integrations), partial / full refund, mark COD collected, resend notifications.
- Returns and exchanges workflow with reason codes and restock option.

5.6 Customers

Profiles, order history, lifetime value, tags, notes, export, and consent status for marketing messages. Staff see only the data their role allows.

5.7 Discounts and content

- Coupons (percentage, fixed, free shipping, buy-X-get-Y), limits, date range, minimum cart, per-customer cap, per-country rules.
- Content manager: hero banners, announcement bar, offer banner with timer, homepage section order, static pages (FAQ, shipping, returns).

5.8 Reports

Sales by day / product / category / phone model / country / channel, GST report (GSTR-ready export), stock valuation, POS cashier and shift report, discount usage, refund report. Export to CSV / Excel / PDF.

5.9 Settings

Store profile, GST details and invoice numbering, currencies and exchange rates, shipping zones, tax rules, payment gateways, notification templates, languages, locations, printers, integrations (Razorpay, Stripe, PayPal, courier, WhatsApp, email).

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6. Roles, permissions and access control

6.1 Principles

- **Least privilege:** staff get only the permissions their job needs.
- The **Owner** can create staff accounts, assign roles, create custom roles and revoke access instantly.
- Every login, permission change and sensitive action is written to an **audit log**.

6.2 Default roles

Permission	Owner	Manager	Inventory	Cashier	Support
View dashboard and reports	✔	✔	Stock only	Own shift only	–

Permission	Owner	Manager	Inventory	Cashier	Support
Add / edit products	✔	✔	Limited (stock, barcode)	–	–
Delete products	✔	–	–	–	–
Change prices	✔	✔	–	–	–
Manage inventory and purchases	✔	✔	✔	–	–
Open POS and create bills	✔	✔	–	✔	–
Give POS discount	✔	Up to limit	–	Up to small limit with PIN	–
Void bill / POS refund	✔	✔ (approval)	–	Request only	–
View and process orders	✔	✔	Pack and ship	–	View
Issue online refunds	✔	✔	–	–	Up to limit
Manage customers	✔	✔	–	–	View
Manage discounts and content	✔	✔	–	–	–
Manage staff and roles	✔	–	–	–	–
Settings, payments, tax	✔	–	–	–	–
View audit log	✔	–	–	–	–

6.3 Authentication and security

- Email + password with strong password rules; **two-factor authentication** (authenticator app or SMS/email OTP), mandatory for Owner and Manager.
- Staff invitation by email with expiring link; forced password set on first login.
- Session timeout (idle 30 minutes, configurable); one-click "log out all devices".
- Rate limiting and temporary lockout after repeated failed logins.
- Optional IP allow-list and shop-device registration for POS.
- All traffic over HTTPS; passwords hashed (Argon2 or bcrypt); secrets in a vault, not in code.
- Cost price and profit margin fields are hidden from roles without permission.
- Custom roles: Owner can tick individual permissions (view, create, edit, delete, export) per module.

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7. POS billing module

A large **"Open POS"** button in the Admin top bar and dashboard opens a full-screen, touch-friendly billing screen that works on desktop, tablet and phone.

7.1 Core billing flow

1. Cashier starts a **shift** (opening cash amount) and selects the store location.
2. Add items by **barcode scan**, SKU, name search, or tapping product tiles (categories and favourites).
3. Edit quantity, apply item or bill discount (permission and limit based), add notes.
4. Attach customer (search by phone / name, or quick-add; walk-in is allowed).
5. Choose payment: cash, UPI (QR displayed), card (external terminal, record reference), split payment, store credit.
6. Generate bill: **GST-compliant tax invoice** with sequential invoice number, HSN codes, tax breakup, store GSTIN, and QR code.
7. Print (58 mm / 80 mm thermal or A4), share by WhatsApp / SMS / email as PDF, or download.
8. Stock is reduced immediately from the selected location, and the sale appears in orders with channel = POS.

7.2 POS features

Feature	Detail
Hold and resume bills	Park several carts and recall them later
Price lookup	Scan to see price and stock without adding
Returns and exchanges	Look up the original bill, select items, refund by original method or store credit, auto-restock
Discounts	Percentage / fixed, coupon codes, manager PIN override above limit
Customer lookup	History, previous purchases, phone model remembered for quick suggestions
Compatibility helper	Cashier selects the customer's phone model and POS shows only compatible items

Feature	Detail
Cash drawer	Cash in / out entries, expected vs counted cash at shift close
End-of-day report	Sales by payment mode, discounts, returns, taxes, cashier-wise totals (Z-report)
Offline mode	Continue billing without internet; bills queue locally and sync on reconnect, with conflict warnings
Hardware support	Barcode scanner (USB / Bluetooth keyboard mode), thermal printer, cash drawer, label printer
Keyboard shortcuts	F-keys for search, discount, payment, hold, new bill

7.3 Controls

- Bills cannot be edited after finalising; use void (with reason and approval) or credit note.
- Invoice numbers are sequential and never reused; voids stay in the audit log.
- Cashiers cannot see cost price or other cashiers' reports.

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8. Technical approach (suggested)

Layer	Suggestion
Frontend (store + admin)	Next.js (React) + TypeScript + Tailwind CSS; PWA support; i18n library with RTL
Backend / API	Node.js (NestJS) or an equivalent; REST or GraphQL; background job queue
Database	PostgreSQL (orders, inventory, catalogue); Redis (cache, sessions, queues)
Search	Meilisearch or Typesense for instant, typo-tolerant model and product search
Media	Object storage + CDN with automatic image resizing

Layer	Suggestion
Payments	Razorpay (India), Stripe and PayPal (international)
Hosting	Managed cloud (region close to customers), CDN, automated backups, staging environment
Observability	Error tracking, uptime monitoring, structured logs
Alternative	Evaluate headless commerce (Medusa, Saleor) to speed up catalogue, cart and order logic, and build the POS and admin extensions on top

8.1 Core data entities

Product · Variant · Category · Phone Brand · Phone Model · Compatibility (variant ↔ model) · Inventory Item · Location · Stock Movement · Customer · Address · Cart · Order · Order Item · Payment · Refund · Shipment · Discount · Staff User · Role · Permission · Audit Log · POS Shift · POS Bill · Currency Rate · Tax Rule · Shipping Zone.

8.2 Non-functional requirements

Area	Requirement
Availability	99.9% uptime for storefront
Performance	Page API responses < 300 ms (p95); search results < 100 ms
Scalability	Handle 10× traffic during sales (autoscaling, caching)
Security	OWASP Top 10 protections, PCI-DSS scope minimised by using hosted payment fields, regular dependency scanning
Data	Daily backups, 30-day retention, tested restore; GDPR-style export and delete of customer data
Browser support	Last 2 versions of Chrome, Safari, Edge, Firefox; iOS Safari and Android Chrome
Analytics	GA4 or privacy-friendly alternative, Meta/Google ads pixels, consent-based

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9. Free icons for the front end

All libraries below are free, open source and allowed for commercial use. Pick **one primary set** for consistency and add Simple Icons only for brand logos.

Library	Licence	Size / style	Best for	Install
Lucide (recommended for storefront)	MIT / ISC	~1,500 clean outline icons, consistent 24px grid and 2px stroke	Storefront UI	`npm i lucide-react`
Tabler Icons (recommended for Admin + POS)	MIT	5,000+ outline icons	Dashboards, admin panels, POS	`npm i @tabler/icons-react`

Library	Licence	Size / style	Best for	Install
Phosphor	MIT	1,200+ icons in 6 weights (thin to duotone)	Brand-style flexibility, highlighted states	`npm i @phosphor-icons/react`
Heroicons	MIT	~300 icons, outline + solid, by the Tailwind team	Tailwind projects	`npm i @heroicons/react`
Simple Icons	CC0	3,000+ flat brand logos	Payment, social and phone-brand logos	`npm i simple-icons`
Iconify	MIT (aggregator)	100+ sets, 200,000+ icons	Searching across all sets in one place	`npm i @iconify/react`

Recommendation: Lucide for the customer storefront (light, friendly, already close to the current look) and Tabler for the Admin Portal and POS (deeper coverage for tables, billing, barcode, printer and inventory icons). Phosphor duotone can be used for large feature illustrations such as the trust strip. Keep stroke width at 1.75–2px and icon size at 20–24px for consistency.

9.1 Suggested icon map

UI element	Lucide (storefront)	Tabler (admin / POS)
Search	`Search`	`IconSearch`
Cart	`ShoppingCart`	`IconShoppingCart`
Wishlist	`Heart`	`IconHeart`
Account	`User`	`IconUser`
Shop by phone	`Smartphone`	`IconDeviceMobile`
Dark / light toggle	`Moon` / `Sun`	`IconMoon` / `IconSun`
Language / country	`Globe`	`IconWorld`

UI element	Lucide (storefront)	Tabler (admin / POS)
Shipping	`Truck`	`IconTruckDelivery`
Secure payment	`ShieldCheck`	`IconShieldCheck`
Returns	`RotateCcw`	`IconArrowBackUp`
Support	`Headphones`	`IconHeadset`
Install app	`Download`	–
Product / inventory	`Package`	`IconPackage`, `IconBoxSeam`
Barcode / scan	–	`IconBarcode`, `IconScan`
POS billing	–	`IconCashRegister`, `IconReceipt`
Print	`Printer`	`IconPrinter`
Staff and roles	–	`IconUsers`, `IconShieldLock`
Reports	–	`IconChartBar`, `IconReportAnalytics`
Discounts	`BadgePercent`	`IconDiscount`
Settings	`Settings`	`IconSettings`

Payment logos (Visa, Mastercard, PayPal, Apple Pay, Google Pay, UPI) and social logos should come from the **Simple Icons** set or each provider's official brand-asset page, since brand logos have their own usage rules even when a library is free. Always keep a copy of each library's licence file in the repository.

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10. Release plan

Phase	Duration (est.)	Deliverables
0. Discovery and design	2 weeks	Final scope, wireframes and high-fidelity designs in Stitch/Figma, design system (colours, type, icons), data model
1. Foundation	3 weeks	Project setup, authentication, roles and permissions, database, CI/CD, staging

Phase	Duration (est.)	Deliverables
2. Admin: catalogue and inventory	4 weeks	Products, variants, phone brands/models and compatibility, categories, media, stock, bulk import
3. Storefront core	5 weeks	Homepage, Shop by Phone, listing, product page, search, cart, checkout, payments (Razorpay), accounts
4. Orders and fulfilment	3 weeks	Order management, invoices, shipping, notifications, returns
5. POS	4 weeks	Billing flow, GST invoices, shifts, returns, printing, offline mode
6. Global layer	4 weeks	Multi-currency, i18n and RTL, Stripe/PayPal, tax and shipping zones, hreflang SEO
7. QA and launch	3 weeks	Security testing, performance tuning, UAT, data migration, soft launch, monitoring

Estimated total: about 28 weeks with a small team (1 designer, 2–3 developers, 1 QA, 1 product owner). Phases 3 and 5 can run in parallel with more developers.

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11. Risks and mitigations

Risk	Mitigation
Stock mismatch between online and POS	Single inventory source, real-time deduction, reservation during checkout, regular cycle counts
Wrong compatibility data causes returns	Mandatory model selection on product creation, compatibility review step, customer "fit" feedback loop
Payment fraud and chargebacks	3-D Secure, gateway risk tools, COD limits and OTP verification
Tax errors across countries	Configurable tax rules, accountant review before enabling each new market

Risk	Mitigation
Internet outage at the shop	POS offline mode with sync and conflict handling
Staff misuse of permissions	Least-privilege roles, approvals, PIN for overrides, audit log, 2FA
Scope growth	Phase gating; anything outside this PRD goes to the backlog

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12. Open questions for the owner

1. Which **countries** are the top 2–3 target markets (decides languages, payment methods and tax setup)?
2. Do you ship internationally yourself or through a courier aggregator (Shiprocket, DHL, others)?
3. Do you have a physical shop today? How many counters and which hardware (barcode scanner, thermal printer)?
4. Are you GST-registered, and do you need B2B invoices with the buyer's GSTIN?
5. Do you want to keep Razorpay only for India and add Stripe/PayPal for the rest?
6. How many staff and what roles will use the Admin Portal at launch?
7. Do you want to keep the orange brand colour, or change the accent?
8. Should the platform be custom-built or based on a headless commerce engine such as Medusa or Saleor?

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13. Acceptance criteria (summary)

- An authorised staff member can add a product with 3 variants, compatibility for 10 models and images in under 3 minutes.
- A shopper selecting "iPhone 17 Pro" sees only compatible products on every page.
- A cashier can scan 3 items, apply a permitted discount, take split payment and print a GST invoice in under 30 seconds, and stock updates on the website immediately.
- A Cashier role cannot open Settings, see cost prices or delete products, and attempts are logged.
- Currency and language switching changes prices and layout (including RTL) without breaking pages.

- Owner can invite a staff member, assign a role, and revoke access with immediate effect.